

who prefer Python over C++ . The resulting neural nets obtained as output from this library can be saved as Python-pickled objects and, hence, can be accessed from Python.

Note that the original version of the project is no longer being developed, but has been reworked into a successor named CUDA-Convnet2. It supports multiple GPUs and Kepler-generation GPUs.

H2O. H2O is an open source machine learning framework developed by Oxdia. H2O's algorithms are basically geared for business processes like fraud or trend predictions. H2O can easily interact in a standalone fashion with different HDFS stores. It can be in MapReduce, on top of YARN or directly in an Amazon EC2 instance.

Hadoop MapReduce can use Java for interaction with H2O, but this framework also provides bindings for R, Python and Scala. It enables cross-interaction with all libraries available on those platforms.

Applications of machine learning

The world is on the path to becoming smarter through automation of all possible manual tasks. Google and Facebook use machine learning to push their respective advertisements to relevant users. Given below are a few applications that you should know of.

Banking and financial services.

Machine learning is widely used to predict customers who are likely to be defaulters in paying credit card bills or repaying loans. This is of utmost importance as machine learning helps banks identify customers who can be given credit cards and loans.

Healthcare. It is widely used to diagnose various deadly diseases (like cancer) on the basis of patients' symptoms, and tallying these with the past data available for similar kinds of patients.

Retail. Machine learning is used to identify products that sell fast and those that do not. It helps retailers decide on the kind of products to

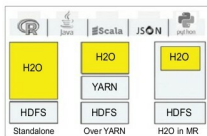


Fig. 4: H2O on Hadoop (Image courtesy: www.infoworld.com)

introduce or remove from their stock. Also, machine learning algorithms can be very effective in finding two or more products that will sell together. This is basically done to encourage customer loyalty initiatives which, in turn, help different retailers develop and maintain loyal customers. Walmart, Amazon, Big Bazaar and other such retail chains extensively make use of machine learning.

Publishing and social media.

There are publishing firms like LexisNexis and Tata McGraw Hill that make use of machine learning to run queries and fetch documents required by their users online, based on their preferences and requirements. Google and Facebook also use these techniques to rank their search outputs and news feeds. Facebook provides a list of possible friends under Friend Suggestions using this.

Robot locomotion. Robot locomotion is a collective term used for the different methods that robots use to transport themselves from one place to other. A major challenge in this field lies in developing capabilities for different robots to autonomously decide how, when and where to move. Machine learning helps them do this quite easily.

Apart from this, there are various decisions that robots need to take instantaneously while they perform activities, which is possible using different machine learning techniques.

Game playing. A strategy game is one in which the player's autonomous decision-making skills are significant in determining the final outcome. Almost all strategy games

require internal decision tree style of thinking, and very high situational awareness. Machine learning meets all these requirements and, hence, is widely used in gaming.

Advantages of machine learning

Machine learning techniques help the system take decisions on the basis of training data in dynamic or uncertain situations.

It can handle multi-dimensional, multi-variety data, and can extract implicit relationships within large data sets in a dynamic, complex and chaotic environment.

It allows reduction of the time cycle and improves resource utilisation. It also provides different tools for continuous quality improvement in any large or complex process.

Another advantage of machine learning techniques is the increased usability of various applications of algorithms due to source programs like Rapidminer. Machine learning allows easy application and comfortable adjustment of parameters to improve classification performance.

Challenges of machine learning

A very common challenge is acquisition of relevant data. Once available data is secured, it often has to be pre-processed depending on the requirements of the specific algorithm used, which has a critical impact on the final results.

Sometimes, interpretation of results also becomes a challenge, as these need to be interpreted according to the algorithm chosen. Different machine learning techniques can be implemented in order to let the system decide on what action it needs to take and when it needs to be taken.

Machine learning can give an edge to automation, and has already helped in making the world a lot smarter. But machines have not stopped learning, and the next level of this technology is being worked on. **EFY**

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